

Total No. of Questions : 8]

SEAT No. :

P7611

[Total No. of Pages : 2

[6180]-130

T.E. (Electronics and Computer Engineering)
SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
(2019 Pattern) (Semester - II) (310352)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain COCOMO model for project estimation with suitable example. [8]
b) What is project scheduling? What are the basic principles of project scheduling. [6]
c) What is task network in project scheduling. [4]

OR

- Q2)** a) Compare Lines of code (LOC) and Function Point (FP) based estimation of software. [8]
b) What is time line chart? Explain with suitable example. [6]
c) What are the categories of software Engineering resources. [4]

- Q3)** a) What are the software design quality attributes and quality guidelines? [7]
b) Explain the design concept: Functional Independence. A design should have high cohesive and low coupling. Justify. [6]
c) Explain guidelines of component level designing. [4]

OR

- Q4)** a) Why is software architecture being important? Explain Data centered Architectural style in detail. [7]
b) Explain in detail call and return architecture style. [6]
c) What is difference between abstraction and refinement. [4]

P.T.O.

- Q5)** a) Explain RMMM plan with suitable example. [8]
b) Explain SCM repository in detail. What are the advantages of SCM repository. [6]
c) What are the different categories of risk? [4]

OR

- Q6)** a) What are the layers of SCM process? Explain each in detail. [8]
b) Explain Risk identification and assessment process for a software project. [6]
c) Write a short note on: Risk table. [4]

- Q7)** a) What is software testing? Explain software testing strategies for software development. [7]
b) How Top down and Bottom up integration is achieved? [6]
c) Compare Alpha and Beta Testing. [4]

OR

- Q8)** a) Explain Defect life cycle in detail. [7]
b) What is test plan, test scenario and test cases. How to write test cases for Login page. [6]
c) Differentiate between verification and validation. [4]



Total No. of Questions : 8]

SEAT No. :

PC1794

[6353]-113

[Total No. of Pages :2

T.E. (Electronics and Computer Engineering)

SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(2019 Pattern) (Semester- II) (310352)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figure to the right indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1) a) What is need of project estimation? What are the steps while estimation of software. [8]
- b) How do you calculate FP and how it is used in estimation of software project? [6]
- c) What are basic principles of project scheduling? [4]

OR

- Q2) a) Compare Lines of code (LOC) and Function Point (FP) based estimation of software. [8]
- b) Explain decomposition techniques. [6]
- c) What is task network in project scheduling? [4]

- Q3) a) Explain data flow architecture style with neat diagram. [7]
- b) Explain the design concept: Functional Independence. A design should have high cohesive and low coupling. Justify. [6]
- c) Explain user interface design issues. [4]

OR

- Q4) a) What are the software design quality attributes and quality guidelines? [7]
- b) Explain component level design for Web Apps. [6]
- c) What is difference between abstraction and refinement. [4]

P.T.O.

- Q5)** a) Explain RMMM plan with suitable example. [8]
b) Explain SCM repository in detail. What are the advantages of SCM repository. [6]
c) Write a short note on: Change control process. [4]

OR

- Q6)** a) What are the layers of SCM process? Explain each in detail. [8]
b) What is risk identification? What are the different categories of risks?[6]
c) Write a short note on: Risk table. [4]

- Q7)** a) Explain defect life cycle in detail. [7]
b) What do you understand by Integration testing? Explain objectives of integration testing. [6]
c) Compare Alpha testing and Beta Testing. [4]

OR

- Q8)** a) What do you understand by System testing? What are the different kinds of system testing that are usually performed on large software? [7]
b) What is test plan, test scenario and test cases? How to write test cases for Login Page? [6]
c) What is GUI testing? Give advantages and drawbacks of GUI testing.[4]



Total No. of Questions : 8]

SEAT No. :

P415

[6003]-503

[Total No. of Pages : 2

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SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
(2019 Pattern) (Semester - II) (310352)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is Software Estimation? What are the Software Estimation Techniques? [9]

b) What is Functional Point Method? Explain with an example. [9]

OR

Q2) a) What is Empirical Estimation model? What are the Empirical Estimation techniques? [9]

b) What is LOC-Based Estimation Method? Explain with an example. [9]

Q3) a) Explain the Software Design Process How to design a Traffic Control System? [8]

b) What is design engineering in software engineering? What are the three types of design in software engineering? [9]

OR

Q4) a) What is Software Architecture? Why is Software Architecture Important? [8]

b) What are Component-Level Design Elements? Explain Component Level Design for Web Apps. [9]

Q5) a) What is software configuration management? What are the 4 main functions of software configuration management? [9]

b) What is the RMMM Plan? Explain in detail. [9]

OR

P.T.O.

Q6) a) Explain software Configuration Management for any suitable software system. **[9]**

b) What is the SCM Repository? Explain in details. **[9]**

Q7) a) What are the different methods of testing? What are the phases involved in Software Testing Life Cycle? **[9]**

b) What are the different levels of testing? What is the difference between functional and non-functional testing? **[8]**

OR

Q8) a) What is Verification and Validation in Software Testing? On what basis the acceptance plan is prepared? **[9]**

b) What is a test case? Explain Bug Life Cycle or Defect life cycle. **[8]**



Total No. of Questions : 8]

SEAT No. :

PB3852

[Total No. of Pages : 2

[6262]-114

T.E. (**Electronics** and Computer Engineering)

SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(2019 Pattern) (Semester - II) (310352)

Time : 2½ Hours]

[Max. Marks : 70

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- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain COCOMO model for project estimation with suitable example.[8]
b) What is project scheduling? What are the basic principles of project scheduling? [6]
c) What is a task network in project scheduling? [4]

OR

- Q2)** a) Compare Lines of code(LOC) and Function Point(FP) based estimation of software. [8]
b) What is a timeline chart? Explain with suitable example. [6]
c) What are the categories of software Engineering resources? [4]

- Q3)** a) List out different design model.Explain Interface design model in detail.[7]
b) Explain the design concept: Functional Independence. A design should have high cohesive and low coupling. Justify. [6]
c) Write a short note on: Modelling Component level design. [4]

OR

- Q4)** a) Why is software architecture being important? Explain Data centered Architecture style in detail. [7]
b) What is design engineering in software engineering? Explain the design concept: Modularity, Refactoring. [6]
c) What is the difference between abstraction and refinement? [4]

P.T.O.

- Q5) a) Explain RMMM plan with suitable example. [8]**
b) Explain SCM repository in detail. What are the advantages of SCM repository? [6]
c) What are the different categories of risk? [4]

OR

- Q6) a) What are the layers of the SCM process? Explain each in detail. [8]**
b) Explain Risk identification and assessment process for a software project.[6]
c) Write a short note on: Risk table. [4]

- Q7) a) What is software testing? Explain software testing strategies for software development. [7]**
b) How Top down and Bottom up integration testing is achieved? [6]
c) Compare Alpha and Beta Testing. [4]

OR

- Q8) a) Explain Defect life cycle in detail. [7]**
b) What is test plan, test scenario and test cases? How to write test cases for Login Page. [6]
c) Differentiate between verification and validation. [4]

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